

5 Interesting Reference Designs Of DC-DC CONVERTERS FOR EVs

Build better EV power systems with these five DC-DC converter reference designs for charging and battery management.

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DC-DC converter reference designs help engineers address the demands of efficient power transfer between battery systems, onboard electronics, and charging infrastructure, including high efficiency, high power density, and bidirectional operation. The five reference designs cover applications ranging from 12V/48V battery management and onboard power conversion to high-power isolated converters for DC fast charging.

60kW isolated DC-DC converter

This reference design uses a dual active bridge topology and fourth-generation silicon carbide power modules for bidirectional power conversion at high efficiency and power density.



It operates at a switching frequency of 100kHz with zero-voltage switching on both sides. The leakage inductance of the high-frequency isolation transformer handles power transfer, eliminating the need for an additional magnetic component. The converter achieves a peak efficiency of 99.2%, maintains efficiency above 98.8% above 20kW, and delivers a power density of 10.5kW/L, including the transformer. The evaluation platform includes isolated gate drivers, an embedded controller with programmable firmware, CAN communication, and protection features such as desaturation-based overcurrent protection, Miller clamping, and temperature monitoring. The modular design allows multiple converters to

be connected in parallel to achieve higher power levels.

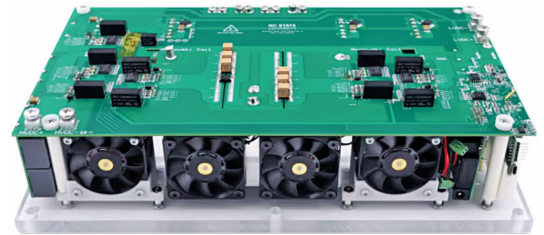
Applications. The design is suitable for renewable energy systems, energy storage, and EV fast-charging infrastructure.

Link. <https://www.electronicsforu.com/electronics-projects/60kw-isolated-dc-dc-converter>

OEM Brand. Wolfspeed

SiC-based polymorphic DC-DC converter

The reference design is a silicon carbide (SiC)-based isolated polymorphic DC-DC converter for 30kW applications. It uses 1200V SiC MOSFETs and SiC Schottky barrier diodes in an isolated asymmetrical dual full-



bridge (phase-shifted full-bridge) topology to achieve 98.5% efficiency and a power density of 7.2kW/L. The converter supports a 4:1 output voltage range while maintaining zero-voltage switching across all operating conditions to reduce switching losses. It uses a carrier-based modulation method similar to that of a buck converter and a proportional-integral (PI) controller to regulate output voltage and current. Software-selectable constant-voltage and constant-current modes are also supported. The reference design allows engineers to evaluate the efficiency, voltage range, and switching performance of SiC devices in high-power DC-DC conversion systems.

Applications. The design is suitable for hybrid and electric vehicle chargers, fast DC chargers, renewable energy systems, industrial